Zachary JC Tobias

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PROFESSIONAL APPOINTMENTS

Postdoctoral Scholar **May 2024 – present**
Department of Ecology, Behavior, and Evolution
University of California - San Diego, La Jolla, CA

Lecturer **August 2025 – present**
School of Biological Sciences
University of California - San Diego, La Jolla, CA

Lecturer II **January 2025 – present**
Department of Biology
University of San Diego, San Diego, CA

EDUCATION

PhD in Biological Oceanography **September 2018 – April 2024**
MIT-Woods Hole Oceanographic Institution Joint Program
Cambridge and Woods Hole, MA
Dissertation: "Genomic and physiological adaptation to temperature in the invasive golden star tunicate, *Botryllus schlosseri*" link

Master of Science in Zoology **December 2015 – December 2016**
University of Otago, Dunedin, New Zealand
Thesis: "Evolutionary genetics of two host-parasite systems on New Zealand's South Island" link
Awarded with Distinction, Mark: A

Bachelor of Arts in Biology **September 2008 – May 2012**
Lewis & Clark College, Portland, OR

TEACHING EXPERIENCE

Instructor of Record **Summer 2026**
BILD 1: The Cell, UC San Diego
- Taught an introductory cell and molecular biology lecture-based course connecting biochemistry to cellular processes, inheritance, and organismal phenotype.
- Emphasized evidence-based reasoning and problem solving over memorization alone.

Instructor of Record **Spring 2025, 2026**
BIOL 242L: Genomes and Evolution Lab, University of San Diego
- Taught two offerings of an introductory evolution and genetics lab using microbiology experiments, simulations, and comparative anatomy to trace the flow of biological information across scales, from cells to populations to lineages.
- Helped students construct scientific arguments by connecting primary literature, study design, and statistical analysis in a final lab report.

Instructor of Record **Summer 2025**
BILD 3: Organismic and Evolutionary Biology, UC San Diego

- Taught an introductory evolution and ecology lecture-based course connecting mechanisms of evolutionary change to organismal diversity, ecological interactions, and global change.
- Helped students connect core concepts to contemporary issues in biodiversity, climate change, scientific racism, and conservation.

Teaching Assistant **Spring 2020**
Biological Oceanography, MIT-WHOI Joint Program

Demonstrator (*Teaching Assistant*) **Fall 2016**
Behavioural and Evolutionary Ecology, University of Otago

Demonstrator (*Teaching Assistant*) **Spring 2016**
Freshwater Ecology, University of Otago

Exam Reader/Writer **Fall/Spring 2016**
Disability Information and Support, University of Otago

Acting Teaching Lab Assistant **Spring 2013 – 2015**
Neurobiology, Lewis & Clark College

CURRICULAR AREAS AND PEDAGOGICAL DEVELOPMENT

Introductory biology; cell and molecular biology; genetics; evolution and ecology; biodiversity; marine biology; biological oceanography; quantitative reasoning and data analysis; scientific writing; inquiry-based laboratory instruction.

Pedagogical training:

Pathways to Scientific Teaching **Winter 2025**
Prof. Diane Ebert-May, UC San Diego.
Certificate seminar for postdoctoral fellows in biology focused on inclusive, learner-centered STEM instruction; evidence-based teaching strategies; active learning; aligned learning objectives, assessments, and instruction; 3-dimensional learning assessment protocol

MENTORING EXPERIENCE

Mikenna Pruette, San Diego State University, STARS Student **Summer 2025**
Mentored project on sexual dimorphism in stickleback; mentee 1st-author manuscript in prep.

Gareth Miller, Western Washington University, WHOI Guest Student **Summer/Fall 2023**
Mentored project on salinity tolerance in a tunicate; mentee 2nd-author manuscript in prep.

Junke "Kelly" Lin, Falmouth Academy **Fall 2022**
Mentored a benthic community ecology project; mentee advanced to state finals.

James Heiser, University of Tampa, WHOI Summer Student Fellow **Summer 2022**
Current PhD student at Scripps Inst. Oceanography; NSF GRFP Recipient.

RESEARCH EXPERIENCE

Postdoctoral Research **May 2024 – present**
Department of Ecology, Behavior, and Evolution, Supervisor: Prof. Diana Rennison
University of California - San Diego
- Investigate determinants of evolutionary parallelism in the threespine stickleback, *Gasterosteus aculeatus*, using genomic, transcriptomic, and morphometric approaches.

Doctoral Research **September 2018 – April 2024**
Department of Biology, Advisor: Dr. Carolyn Tepolt
Woods Hole Oceanographic Institution
- Investigated thermal adaptation and phenotypic plasticity in the invasive tunicate *Botryllus schlosseri* using population genomics and physiological experiments

Lab Volunteer **February 2017 – June 2018**
Funk Lab, Department of Biology
Colorado State University

Master's Research **December 2015 – December 2016**
Department of Zoology, Advisor: Prof. Robert Poulin
University of Otago
- Investigated phylogeny and phylogeography of NZ hairworms (Nematomorpha) and the mermithid nematode *Thaumamermis zealandica*

Microscopy Coordinator **November 2013 – June 2015**
Department of Biology
Lewis & Clark College

Laboratory Technician **June 2012 – June 2015**
Weissman-Unni Lab, Department of Biology
Lewis & Clark College

Summer Research Intern **May 2011 – August 2011**
Jaffe Lab, Marine Physics Laboratory
Scripps Institution of Oceanography, University of California - San Diego

OUTREACH, SERVICE, AND CONSERVATION EXPERIENCE

Science Fair Mentor **September 2022 – February 2023**
Falmouth Academy

Treasurer and Biological Oceanography Representative **October 2020 – September 2021**
MIT-WHOI Joint Program Student Organization

Trails Technician **March – June 2018**
Larimer County Department of Natural Resources, CO

Conservation Easement Monitor and Outreach Specialist **April – October 2017**
Larimer County Department of Natural Resources, CO

PUBLICATIONS (* signifies equal contribution, † signifies mentee author)

Peer-reviewed publications:

ZJC Tobias, G Miller†, CK Tepolt. (2025). Variation of thermal tolerance during northward range expansion in the invasive golden star tunicate, *Botryllus schlosseri*. Conservation Physiology. **13**, coaf018. link

ZJC Tobias, AR Solow, CK Tepolt. (2024). Geography and developmental plasticity shape post-larval thermal tolerance in the golden star tunicate, *Botryllus schlosseri*. Journal of Thermal Biology. **119**, 103763. link

ZJC Tobias, AE Fowler, AMH Blakeslee, JA Darling, ME Torchin, AW Miller, GM Ruiz, CK Tepolt. (2021). Invasion history shapes host transcriptomic response to a body-snatching parasite. *Molecular Ecology.* **30**, 4321–4337. link

L Weston, Z Cook, T Stackhouse, M Sal, B Schultz, **ZJC Tobias**, VR Osterberg, N Brockway, S Pizano, G Glover, VK Unni, and TA Weissman. (2021). In vivo aggregation of presynaptic alpha-synuclein is not influenced by its phosphorylation at serine-129. *Neurobiology of Disease.* **152**, 105291. link

MB Parsley, ML Torres, S Banerjee, **ZJC Tobias**, CS Goldberg, MA Murphy, and MC Mims. (2020). Multiple lines of genetic inquiry reveal effects of local and landscape factors on an amphibian metapopulation. *Landscape Ecology*. **35**, 319–335. link

NL Brockway*, ZT Cook*, M O'Gallagher*, **ZJC Tobias***, M Gedi, KM Carey, YA Pan, MR Metz, and TA Weissman. (2019). Multicolor lineage tracing using *in vivo* time-lapse imaging reveals coordinated death of clonally related cells in the developing vertebrate brain. *Developmental Biology.* **453**, 130-140. link

ZT Cook, NL Brockway, **ZJC Tobias**, J Pajarla, IS Boardman, H Ippolito, S Nkombo-Nkoula, and TA Weissman. (2019). Combining near-infrared fluorescence with Brainbow to identify specific genes within a multicolor context. *Molecular Biology of the Cell*. **30**, 491-505. link

AK Yadav, **ZJC Tobias**, and A Schmidt-Rhaesa. (2018). *Gordionus maori* (Nematomorpha, Gordiida), a new species of horsehair worm from New Zealand. *New Zealand Journal of Zoology*. **45**, 29-42. link

ZJC Tobias, AK Yadav, A Schmidt-Rhaesa, and R Poulin. (2017). Intra- and interspecific genetic diversity of New Zealand hairworms (Nematomorpha). *Parasitology*. **144**, 1026-1040. link

ZJC Tobias, F Jorge, and R Poulin. (2017). Life at the beach: comparative phylogeography of a sandhopper and its nematode parasite reveals extreme lack of parasite mtDNA variation. *Biological Journal of the Linnean Society*. **122**, 113-132. link

M Marra, **ZJC Tobias**, HR Cohen, G Glover, and TA Weissman. (2015). *In vivo* time-lapse imaging in the zebrafish lateral line: a flexible, open-ended research project for an undergraduate neurobiology lab course. *Journal of Undergraduate Neuroscience Education*. **13**, 215-224. link

KR Hamling, **ZJC Tobias**, and TA Weissman. (2015). Mapping the development of cerebellar Purkinje cells in zebrafish. *Developmental Neurobiology*. **75**, 1174-1188. link

Preprints and manuscripts in progress:

ZJC Tobias, A Härer, and DJ Rennison. (submitted). Evolutionary constraint predicts the repeatability of transcriptional divergence during repeated adaptation. Preprint link

ZJC Tobias and CK Tepolt. (submitted). A distributed genomic architecture of repeated thermal adaptation in a low-dispersal marine invader. Preprint link

T Everts, TL Schmidt, **ZJC Tobias**, A Bierdrzycka, AR Sousa, A Kouba, C Wang, PJ Haubrock. (submitted). Where and when do invaders adapt? A population-level framework for adaptation during biological invasions.

M Pruette†, **ZJC Tobias**, and DJ Rennison. (in prep). Contrasting patterns of sexual dimorphism across lakes during early stages of speciation in the threespine stickleback.

M Acord, Gareth Miller†, CK Tepolt, **ZJC Tobias**. (in prep). Low salinity tolerance and recovery of the invasive tunicate, *Botrylloides violaceus.*

FELLOWSHIPS, GRANTS, AND AWARDS

JiJi Foundation, *PI. Co-PI: Gorgonio Ruiz Campos, UABC Ensenada* **2027**
"Ecological monitoring and conservation genomics of the last Mexican population of the threespine stickleback (*Gasterosteus aculeatus*)"
Value = $19,610 USD

North Pacific Research Board Core Program, *co-authored* **2022-2024**
"An evolving threat: thermal adaptation and the spread of an invasive biofouling tunicate in the Gulf of Alaska"
Value = $211,734 USD

Woods Hole Oceanographic Institution Grassle Fellowship Fund **2022**
Value = $7,500 USD

American Philosophical Society Lewis and Clark Field Scholar **2021**
Value = $4,500 USD

National Defense Science and Engineering Graduate Fellowship **2020 – 2023**
Est. value = $333,950 USD

Woods Hole Oceanographic Institution Ocean Ventures Fund **2019**
Value = $6,000 USD

University of Otago International Research Master's Scholarship **2016**
Value = $37,833 NZD

University of Otago Department of Zoology Conference Grant **2016**
Value = $300 NZD

New Zealand Society of Parasitology Student Travel Grant **2016**
Value = $500 NZD

Aquaneering Art of Science Imaging Contest, 1st Place Winner. **2015**
Value = $1,000 USD

Lewis & Clark College Dean's Scholarship **2008 – 2012**
Value = $12,000 USD ($3,000 per annum)

POSTERS AND PRESENTATIONS († signifies undergraduate mentee presenter)

ZJC Tobias, M Vivanco-Berkovich, A Ferreira-Arietta, R Muñiz-Salazar, JM Sandoval-Gil. Examining the transcriptomic basis of physiological responses to a simulated marine heat wave in the surfgrass *Phyllospadix scouleri*. Western Society of Naturalists. November 15th, 2026. Accepted for oral presentation.

MM Pruette†, C Wise, **ZJC** Tobias, DJ Rennison. Contrasting patterns of sexual dimorphism during parallel ecotypic divergence in threespine stickleback. Society of Naturalists. November 13th, 2026. Accepted for oral presentation.

ZJC Tobias, BD Gastfriend, MC Farnitano, M Tresguerres, R Daneman, DJ Rennison. Cell-type-resolved regulatory evolution during marine–freshwater divergence in stickleback. Population, Evolutionary, and Quantitative Genetics. June 11th, 2026. Poster presentation.

MM Pruette†, **ZJC Tobias,** DJ Rennison. Exploring patterns of sexual dimorphism in sympatric ecotypes of stickleback. Western Society of Naturalists. November 9th, 2025. Poster presentation.

ZJC Tobias and DJ Rennison. Exploring parallelism of gene expression across ecotypes of the threespine stickleback (*Gasterosteus aculeatus*). Western Society of Naturalists. November 9th, 2025. Oral presentation.

ZJC Tobias. Exploring the molecular basis of parallel evolution in tunicates and stickleback. UCSD Ecology, Behavior, and Evolution Seminar. November 22nd, 2024. Oral presentation.

ZJC Tobias, G Miller, CK Tepolt. Plasticity and adaptation shape geographic variation of thermal tolerance in the invasive golden star tunicate, *Botryllus schlosseri*. Western Society of Naturalists. November 8th, 2024. Oral presentation.

ZJC Tobias and CK Tepolt. Population genomics of an invasive tunicate reveals limited parallelism during local adaptation to temperature across both coasts of North America. The Evolution Meeting. June 28th, 2024. Oral presentation.

ZJC Tobias, G Miller, CK Tepolt. Differentiation of thermal tolerance in the invasive golden star tunicate, *Botryllus schlosseri*. Alaska Marine Science Symposium. January 29th, 2024. Poster presentation.

G Miller†, CK Tepolt, **ZJC Tobias**. Do Southeast Alaska salinities prevent an invasive barrier to *Botrylloides violaceus*? Alaska Marine Science Symposium. January 29th, 2024. Poster presentation.

ZJC Tobias. Geography and developmental plasticity shape post-larval thermal tolerance in the golden star tunicate, *Botryllus schlosseri*. WHOI BioSeminar. August 17th, 2023. Oral presentation.

ZJC Tobias and CK Tepolt. Whole-genome data reveal extremely fine-scale population structure in *Botryllus schlosseri.* International Conference on Marine Bioinvasions XI. May 19th, 2023. Oral presentation. (*Presented by CKT, as I was on paternity leave*).

CK Tepolt and **ZJC Tobias**. Understanding marine invasive species with genomics and ecophysiology. Alaska Invasive Species Partnership. October 11th, 2022. Oral presentation.

ZJC Tobias. Low-coverage whole genome sequencing reveals multiple introductions of a highly invasive tunicate to the west coast. NDSEG Conference. July 28th, 2022. Oral presentation.

ZJC Tobias. How to make a zombie: coordinated changes in gene expression in response to a body-snatching parasite. WHOI Student BioSeminars. February 27th, 2020. Oral presentation.

ZJC Tobias, GM Ruiz, and CK Tepolt. Assessing the transcriptional response to an invasive body-snatching parasite in the mud crab *Rhithropanopeus harrisii*. The Evolution Meeting, June 22nd, 2019. Poster presentation.

ZJC Tobias and R Poulin. Evolutionary genetics of New Zealand hairworms (Nematomorpha), *Thaumamermis zealandica* (Mermithidae), and their hosts. University of Otago Zoology Postgraduate Symposium, November 11th, 2016. Oral presentation.

ZJC Tobias and R Poulin. Comparative population genetics in two host-parasite systems of New Zealand's South Island. New Zealand Society of Parasitology Annual Meeting, October 27th, 2016. Oral presentation.

ZJC Tobias, YA Pan, and TA Weissman. Multicolor lineage tracing of neuronal progenitors in the larval zebrafish hindbrain. Society for Neuroscience Oregon Chapter Meeting, April 5th, 2014. Poster presentation.

REVIEWER

Molecular Ecology (2x)*, Ecography* (2x), *Philosophical Transactions of the Royal Society B*, *Journal of Parasitology*, *Annals of Applied Biology, Biological Invasions* (2x)*, Zootaxa,* Graduate Women in Science Fellowship, *Diversity and Distributions* (2x), *Journal of Helminthology*, *Biological Bulletin, Proceedings of the Royal Society B*, *Marine Ecology Progress Series*

REFERENCES

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